

CHATSWORTH DATA CORPORATION

Technical Manual

For

Optical Mark Reader 1102

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1. SCOPE

This specification describes the features and performance characteristics of the Optical Mark Reader 1102 manufactured by Chatsworth Data Corporation. Instructions for installing, testing, and programming the OMR-1102 are described in detail in the following sections.

2. GENERAL INFORMATION

2.1 Description

The OMR-1102 is a programmable, microprocessor-based Optical Mark Reader. The OMR is designed to scan standard 12-channel, 3.25 inches wide response documents and convert the marked data to an ASCII character format that can be transmitted to a computer via a serial data communications port (RS232C). This alternative to keyboard allows for quick data entry.

2.2 Response Documents

There are numerous document designs for use on the OMR-1102.

A basic card utilizes a maximum of 12 data channels (defined as 12 rows which run the length of the document). There are up to 126 vertical columns available over the length of the document. Though there is no maximum document length, practical considerations limit the document length to between 7 3/8 and 14 inches. A minimum of three inches is required to transport the document. Each vertical column is defined with a timing mark along the left edge of the card to define the portion of the response area that will be scanned for the presence of a mark.

Several card samples are included with your OMR and a more complete discussion of document design, tips, suggestions and specifications are included in Appendix A.

2.3 Serial Communications

The OMR-1102 utilizes a standard serial communication mode defined as RS232C. This communication protocol will support baud rates from 300 to 38400 on any installed communication port.

The data string format is defined as one start bit, seven data bits, even parity, and one stop bit.

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The baud rate default is 9600 and is adequate for nearly all applications and computers. However, if necessary, the baud rate may be modified after power up utilizing the utility program OMRCFG.EXE. See Section 8 for instructions.

See Appendix B for factory defaults.

3. INSTALLATION AND CHECKOUT

3.1 Installation

Before using the OMR-1102 proceed as follows:

Plug the three-prong end of the power cord from the transformer into a 110 VAC outlet. Insert the round male end of the power cord into the power receptacle on the right side of the reader. Next, attach the male end of the data cable to the 25-pin female connector on the right side of the OMR. Attach the female end of your data cable to the communications port on the computer.

If there is a serial “mouse” attached to your computer it will most likely be using the COM1 port. The remaining serial ports should be COM2, COM3 etc. You can identify which communications port the OMR-1102 is attached to by using the utility program OMRDETECT discussed in the Checkout Section.

Note: There are three different data cables available from Chatsworth Data:

PC cable uses: 25-pin male and female connectors

AT cable uses: 25-pin male and 9-pin female connectors

Macintosh cable uses: 25-pin male and 9-pin DIN female connectors

If you did not receive the correct cable with your OMR-1102, contact Customer Service for a replacement. DO NOT attempt to use a standard serial cable. The cable supplied by Chatsworth Data is custom made specifically for use with the OMR-1102.

The ON/OFF switch is also located on the right side of the OMR. Press down on the side of the switch with the small white dot to apply power.

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3.2 Checkout

To insure the unit is powered up take one of the sample response forms and insert it into the throat of the OMR. The motors should turn on and transport the card through the OMR. If there is no response, check the power hookups for secure connections and the AC receptacle for power. If there is still no response, contact Customer Service for assistance.

To check the installation, the utility program OMRDETECT.EXE will examine the system configuration to determine which communications port is being used. It will also determine the current baud rate, and if possible, determine the proper data cable being used. This information is displayed on the screen after the process is complete.

To load the utility program, proceed as follows:

At the C:\> Type [A:] or [B:] and press [Enter].
At the A:\> Type [OMRDETECT.EXE]
Press [Enter]

Assuming the OMRDETECT program executed satisfactorily, the utility program called OMRDEMO will confirm the OMR-1102 is reading properly. If the program is unable to communicate with the OMR, recheck the data cable connections and power source. If the problem still exists, contact Customer Service.

You may use the sample response forms to test the unit. Make several marks in the response “bubbles” with pencil. It helps if you make your marks in a pattern (i.e., mark the A in question 1, the B in question 2, the C in question 3, etc.).

At the A:\> Type [OMRDEMO]
Press [Enter]

The screen will clear and a graph form will display the rows of the card vertically and the columns of the card horizontally. At the prompt insert the card you just marked. After the card has been transported, an image of the marks will be displayed. The image displayed should reflect the marks made on the response form. If some of the marks do not display, examine the marks to make sure they completely fill the “bubble” marking area. This is especially important if the marking area is an oval and the mark was made with a circular motion leaving the center of the oval clear. Correct any mis-marked areas and

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rescan the document. If there still seems to be a problem, contact Customer Service for assistance.

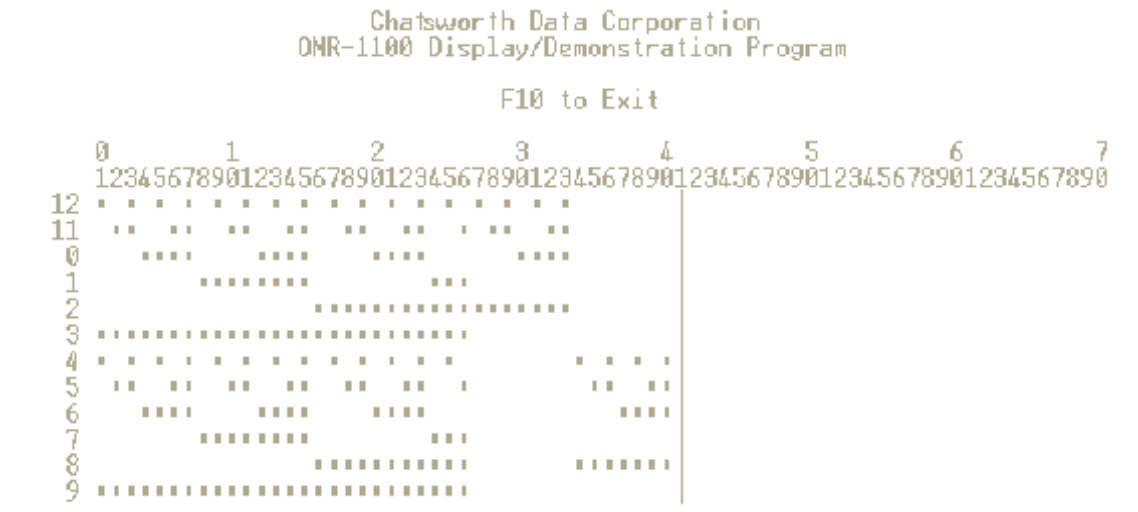
Upon successful completion of the checkout utilities, the OMR-1102 is ready to fulfill the document scanning requirements of your application program.

Note: If the document feed control option was ordered for the OMR, a document will not transport unless the OMR is under the control of a computer application program. Consequently, you will not be able to run the checkout procedures as defined in this section.

An example of a typical display is shown in Figure 1.

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FIGURE 1



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4. DISTRIBUTION SOFTWARE

4.1 Chatscore

Chatsworth Data Corporation developed *Chatscore* as a stand-alone teacher-made scoring test and survey package for use by educational institutions and industry. This application supports scoring of multiple-choice or true and false tests. This program supports over 25 different test scoring card formats and also prints a series of reports based on the response forms processed. *Chatscore's* survey module supports up to 100 multiple-choice responses and produces an analysis for each question.

A complete operator's guide for *Chatscore* can be printed or viewed by running the CSMANUAL.EXE program included on the distribution disk.

To install *Chatscore*, create a subdirectory of your choice (example: CHATSCOR) and install the system using the INSTALL.EXE program as follows:

```
C:\>[MD\CHATSCOR] [Enter]
C:\>[CD\CHATSCOR] {Enter}
C:\CHATSCOR>[INSTALL.EXE] [Enter]
```

The *Chatscore* is comprised of the following programs and files:

CSMANUAL.EXE	Print or Display <i>Chatscore</i> Manual
INSTALL.EXE	Install <i>Chatscore</i> to Hard Drive
SETUP5.EXE	Setup Program
SETUP5.DES	Configuration File
BRUN45.EXE	QB Run-Time Module
CS_EXEL2.EXE	Excelsior Grade Export

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4.2 OMR-1102 Utility Programs

Utility programs are on the distribution disk. These programs assist in setup and checkout of the OMR-1102 and in utilizing some of the advance features.

To install the OMR utility programs create a subdirectory of your choice (example: OMRUTIL) and copy the utility program as follows:

```
C:\>[MD\OMRUTIL] [Enter]
C:\>[CD\OMRUTIL] [Enter]
C:\OMRUTIL>[Copy A:OMR*>* C] [Enter]
```

To facilitate technical support, it is highly recommended the utility programs be installed on any computer that may use the OMR-1102.

The utility program files are:

OMRCFG.EXE:	Configuration Program
OMRCFG.CFG:	Configuration File
OMRDEMO.EXE	Card Image Display
OMRDETECT.EXE	Detect OMR Com: Port, including baud rate and cable information
OMRDNLD.EXE	OMR Universal Driver

For complete instructions on the utility program, see Section 8.

5. DATA MARK CONVERSION

The translation download feature of the OMR-1102 allows a series of commands to be sent via the serial communication port to instruct the OMR how to translate the response form marks into ASCII characters.

Download commands are stored in a volatile memory and are only active as long as the OMR-1102 is powered up. For this reason, it is recommended the appropriate download strings be included in the application program when required.

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Response marks conversion to data may utilize any of four internal formats generated by OMR logic:

Hollerith to ASCII Conversion
Binary to ASCII Conversion (factory default)
Translate Conversion
Split Translate Conversion

In addition to the four conversion modes, a fifth download command called Row Enable is used in conjunction with any of the translation modes to specify which rows on the response form are to be considered for translation.

5.1 Hollerith To ASCII Conversion

Hollerith was used extensively with early computers as the layout defined for punched card. In Hollerith format each vertical column is defined with 12 keypunch or marking areas. The combination of marks or punches can define the standard 128 character ASCII set. The 12 channels are defined in two categories. The top three channels are called zone channels and the bottom nine channels are called field channels.

The zone channels are used alone or in combination with field channels (multiple marks or punches per column) to represent alphabetical and special characters. For definition purposes the top zone is called the “12” channel or “&” channel. The second zone channel is the “11” channel or “-” channel and the third zone channel is the “0” channel.

Field channels are 1 to 9 respectively from top to bottom.

<i>Row</i>	<i>Type</i>	<i>Char</i>
12	Zone	&
11	Zone	-
0	Zone	0
1	Field	1
2	Field	2
3	Field	3
4	Field	4
5	Field	5
6	Field	6
7	Field	7
8	Field	8
9	Field	9

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Utilizing Hollerith to ASCII mode each column of marks will be converted to its equivalent ASCII character translated with one character per column. See [Appendix C](#).

5.2 Binary To ASCII Conversion

Binary coding utilizes two 8-bit bytes per column to determine marked channels in a given column. This conversion mode transmits the entire card to an image format giving each mark in a column specific meaning.

This mode is commonly used for complex layouts with multiple marks within a single column or in a form design that contains a layout not applicable to any other mode.

In Binary mode every marking location on the document (max 12 by 126) can be utilized to represent specific data. The application program controls conversion of a mark to a specific variable.

The Binary Table is shown in [Appendix D](#).

5.3 Translate Conversion

The Translate Conversion mode generates a single character per column that is specified in a mask and downloaded to the OMR. This mode allows 12 characters to be designated to represent the mark in its respective column. In addition, a character may be specified to indicate “no mark” in a column and one to specify more than one mark was made in a column.

Download commands and mask format are outlined in Section 6.

5.4 Split Translate Conversion

Split Translate Conversion mode generates a single character per column for each half of the card. Up to six characters may be defined to represent the right and left sides of the cards. Two additional characters are used to represent no mark exists in the column area, and one additional character to specify more than one mark exists in a column area.

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5.5 Combining Conversion Modes

Most response forms require different layouts within the form design in order to communicate the necessary information. Four conversion modes may be used in combination to complete the translation of the entire card. It should be noted that each mode may be used only once for each card definition.

If a specific card layout cannot be translated in this way, use the Binary to ASCII mode and translate the marks within the application program. The sample test cards included with the OMR are a good example. The top portion of the card contains a student I.D. number that can be translated using the Translate Conversion mode with a numeric mask for 1 through 9. The body of the form can be translated using Split Translate Conversion mode and a mask of A through E to process the multiple-choice questions.

5.6 Row Enable Command

The Row Enable command can be applied to each conversion mode to disable channels not to be used for data. This reduces the risk of double marks in a data string that could be caused by stray marks, doodling, etc. Row Enable is also useful when used to disable a channel or channels to indicate the question number or other pre-printed information not to be translated.

6. CONFIGURATION DOWNLOAD

There are specific download commands to enable each of the various modes and a specific format for each command. Every command must start with a Control R (Decimal 18) followed by the command format. Multiple commands may be transmitted at one time. The entire download command string is terminated with the End Of Transmission ("E") command. The computer to OMR communication must be executed only when the card reader is in an idle state (forms are not being scanned and transport is not running).

The standard download format is:

Ctrl R | Command String | Ctrl R | Command E

When a command download string has been accepted and processed by the OMR, the character "G" followed by a <CR> will be transmitted by the OMR to the host computer. If a command in the download string is NOT executable, the download string, up to and including the character that caused the error, followed by "?" and a <CR> will be transmitted by the OMR to the host computer.

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6.1 Download Commands

There are eight (8) download commands accepted by the OMR:

- End of Transmitted Message ("E")
- Hollerith to ASCII Mode ("H")
- Binary (or image) to ASCII Mode ("I")
- Load Split Translate Conversion Table ("L")
- Row Enable ("R")
- Split Translate Conversion Mode ("S")
- Translate Conversion Mode ("T")
- Load Translate Conversion Table ("X")

6.1.1 End of Transmission (Command "E")

Action: Command "E" is defined as the End of Transmission command

Syntax: Chr\$(18); "E"

Remarks: All download strings transmitted to the OMR must be terminated with Command "E". The transmission of only a Command "E" will clear the card reader of all previous commands that set the mode of operation, i.e., disables all the conversions modes (Commands H, I, S, and T). Transmission of command "E" only will not clear previously loaded translation tables or masks (i.e. commands "L", "R", and "X").

Example: The following code will download the OMR on communication port 1, clearing all conversion mode commands.

```
Open COM1:9600,E as #1  
Print #1, Chr$(18); "E"
```

Output Returns the character "G" followed by a <CR> for a successful download

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6.1.2 Hollerith To ASCII Mode (Command “H”)

- Action: Command “H” sets the OMR in Hollerith to ASCII mode.
- Syntax: Chr\$(18);”H##”
Where ## is the number of columns to be translated (1 to 99).
“00” is used to translate all remaining columns up to a total of 126.
- Remarks: “H” is the character accepted by the OMR to translate marks to Hollerith characters. An invalid combination of marks, which has no associated character in the Hollerith Table, will return a “?”.
- Example: The following command will place the OMR in Hollerith to ASCII mode for 25 columns:
- ```
Open COM1:9600,E as #1
DwnLd$=Chr$(18)+”H25”+Chr$(18)+”E”
Print #1, DwnLd$
```
- Output Returns the character “G” followed by a <CR> for a successful download. When a card is scanned, one ASCII character per column will be returned followed by a <CR>. The characters returned will be based on specific channels marked using the 128-character Hollerith Table (See [Appendix C](#)).

### **6.1.3 Binary To ASCII Mode (Command “I”)**

- Action: Command “I” sets the OMR in Binary to ASCII mode that will return 2 bytes per column.
- Syntax: Open COM1:9600,E as #1  
Chr\$(18);”I##”  
Where ## is the number of columns to be translated (1 to 99).  
“00” is used to translate all remaining columns up to a total of 126.

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Remarks: "I" is the character accepted by the OMR to translate all marks in a column into two bytes. The marks on the right side of the card (rows 12, 11, 0, 1, 2, 3) will be contained in the first byte. The marks on the left side of the card (rows 4, 5, 6, 7, 8, 9) will be contained in the second byte.

Example: The following command will place the OMR in the Binary to ASCII mode for ALL columns:

```
COM1:9600,E as #1
DwnLd$=Chr$(18)+"I00"+Chr$(18)+"E"
Print #1, DwnLd$
```

Output Returns the character "G" followed by a <CR> for a successful download. A mark is reflected as one bit in the byte and 0 bit is no mark. The structure of each byte is made up of 8 bits. The right hand bit is the least significant bit (LSB) and the 7<sup>th</sup> bit is the most significant bit (MSB). The 8<sup>th</sup> bit is the parity bit. The 6<sup>th</sup> bit is forced to a 1 so that a blank column will have a specific value of 32. When a card is scanned, two bytes per column will be returned followed by a <CR>. The characters returned will be in the range of 32 (blank) to 127 (all 12 channels) marked.

The bits are valued from 1 to 64 from the LSB to the MSB as follows:

***Byte 1:***

| Row | Value        |
|-----|--------------|
| 12  | 1 LSB        |
| 11  | 2            |
| 0   | 4            |
| 1   | 8            |
| 2   | 16           |
|     | 32 Always On |
| 3   | 64 MSB       |

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**Byte 2:**

| Row | Value        |
|-----|--------------|
| 4   | 1 LSB        |
| 5   | 2            |
| 6   | 4            |
| 7   | 8            |
| 8   | 16           |
|     | 32 Always On |
| 9   | 64 MSB       |

The 7<sup>th</sup> bit is always ON giving each byte a minimum value of 32. This eliminates transmitting any special or control characters to the host computer that have values less than 32.

**6.1.4 Load Split Translate Conversion Table (Command “L”)**

Action: Command “L” defines the table for the Split Translate Conversion mode.

Syntax: Chr\$(18);"L#####"  
Where ##### are the six (6) characters used in the translation of the marks, plus one (1) for missing mark and one (1) for double mark. The characters are used for both sides of the split response form. Any of the ASCII characters from 33 to 127 may be used.

Remarks: “L” is the character used to set up a conversion table or mask to identify characters to be used with Command “S” (Split Translate). The first argument is the character used to indicate “no” mark, the next 6 arguments are the characters used for the 6 data channels, and the 8<sup>th</sup> argument is the character used for double marks within the 6 data channels. See the table below.

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| Arg. # | Channel |
|--------|---------|
| 1      | NM      |
| 2      | 12 & 4  |
| 3      | 11 & 5  |
| 4      | 0 & 6   |
| 5      | 1 & 7   |
| 6      | 2 & 8   |
| 7      | 3 & 9   |
| 8      | DM      |

NM = No Mark

DM = Double Mark

Example: The following Basic code will download a Split Translate Conversion Table for a multiple-choice card using “\*” for no mark and “#” for multiple mark. The character “x” is used to identify unused channels. In this example channels 3 and 9 are used for the question number and will be disabled with Command “R”.

```
Open COM1:9600,E as #1
E$ = Chr$(18)
DownLd$=E$+"L*EDCBAX#" +E$+"E"
Print #1, DwnLd$
```

Output Returns the character “G” followed by a <CR> for a successful download. When a card is scanned, two ASCII characters per column will be returned followed by a <CR>. The characters returned will be based on the relationship of the data mark channel relative to the conversion table, i.e. a mark in channel 8 on the lower or left-hand side of the split card would return an “A”; a mark in channel 0 in the upper or right-hand side of the split card would return a “C” based on the conversion table used in the example.

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### **6.1.5 Row Enable (Command “R”)**

**Action:** Command “R” is defined as the Channel Select command. This command disables or enables specific channels. Disabled channels are “masked” out, therefore marks in a disabled channel are not translated.

**Syntax:** Chr\$(18);”R####”  
Where #### each of the four arguments made up of a summed Octal value of three (3) channels each.

**Remarks:** Each argument is determined by adding the three values associated with the four groups of channels. The first argument is comprised of channels 9, 8, and 7. The second group uses channels 6, 5, and 4. The third group includes channels 3, 2, and 1. The last group is made up of channels 0, 11, and 12.

Each argument can have a value of 0 to 7 depending on which channels are to be enabled or disabled. An enabled channel is shown as a 1 and a disabled channel as 0. Channels are assigned a value of 1, 2, or 4 (Octal Coding) from right to left if they are to be enabled. If they are to be disabled, a value of 0 is used. To determine the value of the argument, add the three values together.

The following table illustrates the possible combinations to determine the argument values for each group of channels:

|              |     |     |     |     |     |     |     |     |
|--------------|-----|-----|-----|-----|-----|-----|-----|-----|
| <i>Octal</i> | 421 | 421 | 421 | 421 | 421 | 421 | 421 | 421 |
| <i>Group</i> | 000 | 001 | 010 | 011 | 100 | 101 | 110 | 111 |
| <i>Value</i> | 0   | 1   | 2   | 3   | 4   | 5   | 6   | 7   |

1 = Enable Channel

0 = Disable Channel

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Example: The following Basic Code will enable channels 12, 11, 0, 1, 3, 5, 7, and 8.

| Arg. #            | #1      | #2      | #3      | #4        |
|-------------------|---------|---------|---------|-----------|
| Channel           | 9, 8, 7 | 6, 5, 4 | 3, 2, 1 | 0, 11, 12 |
| Enable/Disable #2 | 011     | 010     | 101     | 111       |
| Value             | 3       | 2       | 5       | 7         |

```
Open COM1:9600,E as #1
DwnLd$=Chr$(18)+"R3257"+Chr$(18)+"E"
Print #1, DwnLd$
```

Output Returns the character "G" followed by a <CR> for a successful download. This command does not affect the output data string.

#### **6.1.6 Split Translate Conversion Mode (Command "S")**

Action: Command "S" sets the OMR to Split Translate Conversion mode.

Syntax: Chr\$(18);"S##"  
Where ## is the number of continuous columns to be translated (1 to 99). "00" is used to translate all remaining columns up to a total of 126.

Remarks: "S" is the character accepted by the OMR to translate marks using the Split Translate Conversion layout. This requires the portion of the response form to be translated be designed with two columns of up to 6 responses. This layout has the effect of splitting the card into two halves laid out lengthwise of the document.

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Example: The example opens COM1: for output and downloads the Split Translate Conversion command for 25 columns and a Split Translate Conversion Table (Command “L”).

```
Open COM1:9600,E as #1
E$ = Chr$(18)
E$+"S25"+E$+"L*EDCBAX##"+E$+"E"
Print #1, ChrOut$
```

Output Returns the character “G” followed by a <CR> for a successful download. When a card is scanned two ASCII characters per column will be returned followed by a <CR>. The characters returned are based on the relationship of the data mark to the associated character with that channel in the conversion table (i.e., marks in channels 8 and 0 would return an “A” and “C” respectively using a typical multiple-choice response form).

### **6.1.7 Translate Conversion Mode (Command “T”)**

Action: Command “T” sets the OMR to Translate Conversion mode.

Syntax: Chr\$(18)”T##”  
Where ## is the number of continuous columns to be translated (1 to 99). “00” is used to translate all remaining columns up to a total of 126.

Remarks: Command “T” in conjunction with Command “X” gives the host computer the capability of defining the character to be associated with each of the 12 channels on a response form.

Example: In the Basic Code example, the first 9 columns translate to a student I.D. number. Where the number goes from 0 to 9, “\*” is used to indicate no mark, and “#” is used to indicate multiple marks. Channels 8 and 9 will be “masked” out with Command “R.”

```
Open COM1:9600,E as #1
E$ = Chr$(18)
ChrOut$=E$+"T09"+E$+"X*9876543210xx#"+E$+"E"
Print #1, ChrOut$
```

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**Output** Returns the character “G” followed by a <CR> for a successful download. When a card is scanned one ASCII character per column will be returned followed by a <CR>. The characters returned will be based on the relationship of the data mark to the character associated with that channel in the conversion table (i.e., a mark in channel 6 would return a “1” and a mark in channel 12 would return a “9”).

**6.1.8 Load Translate Conversion Table (Command “X”)**

**Action:** Command “X” defines a conversion table for use with Command “T”.

**Syntax:** Chr\$(18);"X#####"  
Where ##### are the twelve (12) characters used in the translation of marks plus one (1) for missing mark and one (1) for double mark. Any of the ASCII characters from 32 to 127 may be used.

**Remarks:** “X” is the character accepted by the OMR to set up a conversion table to be used in conjunction with Command “T” (Translate Conversion mode). The first argument is the character to be used to indicate no mark, the next 12 arguments are the characters for the 12 data channels, and the 14<sup>th</sup> argument is the character to be used for double marks within the 12 data channels.

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The following table illustrates the relationship between the arguments and data channel, or marks:

| <i>Arg. #</i> | <i>Channel</i>          |
|---------------|-------------------------|
| 1             | No mark character       |
| 2             | 12                      |
| 3             | 11                      |
| 4             | 0                       |
| 5             | 1                       |
| 6             | 2                       |
| 7             | 3                       |
| 8             | 4                       |
| 9             | 5                       |
| 10            | 6                       |
| 11            | 7                       |
| 12            | 8                       |
| 13            | 9                       |
| 14            | Multiple mark character |

Example: The following Basic Code will set the OMR to Translate Conversion mode for the first 9 columns of the response form and load a Translate Conversion table for numeric translations of 0 through 9, "\*" for no mark and "#" for multiple marks. The character "x" is used to identify unused channels. In this example, channels 9 and 8 are used for the hand-written I.D. number.

```
Open COM1:9600,E as #1
E$ = Chr$(18)
ChrOut$=E$+"T09"+E$+"X*9876543210xx#"+E$+"E"
Print #1, ChrOut$
```

Output Returns the character "G" followed by a <CR> for a successful download. When a card is scanned one ASCII character per column will be returned. When the entire response form has been processed, the data string will be terminated with a <CR>. The characters returned are based on the relationship of the data mark channel to the character associated with the specific channel in the Translate Conversion table.

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## **6.2 Combining Modes**

Most forms require more than one command for complete translation of all marks on the card to ASCII data. The following command sequences are related to specific response forms identified as follows:

**Form 1:** Form No. 05-SD is a 100-question multiple-choice response form with a 9 column student I.D. area.

```
Open COM1:9600,E as #1
E$ = Chr$(18)
DL1$ = E$+"R3777"
DL2$ = E$ + "T09"
DL3$ = E$ + "X*9876543210xx#"
DL4$ = E$ + "R3737"
DL5$ = E$ + "S00"
DL6$ = E$ + "L*EDCBAX#"
DL7$ = E$ + "E"
DwnLd$=DL1$+DL2$+DL3$+DL4$+DL5$+DL6$+DL7$
Print #1, DwnLd$
```

**Form 2:** Form No. H55070-0 is a 35-question response form with a 7 column I.D. The multiple-choice questions (A-E) are double spaced across the card.

### ***Example 1 - Using Binary to ASCII Conversion***

```
Open COM1:9600,E as #1
E$ = Chr$(18)
DL1$ = E$+"R1777"
DL2$ = E$ + "T07"
DL3$ = E$ + "X*9876543210xx#"
DL4$ = E$ + "R1252"
DL5$ = E$ + "I00"
DL6$ = E$ + "L*EDCBAX#"
ChrOut$=DL1$+DL2$+DL3$+DL4$+DL5$+DL6$
Print #1, ChrOut$
```

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*Example 2 - Using Translate Mode*

```
Open COM1:9600,E as #1
E$ = Chr$(18)
DL1$ = E$+"R1777"
DL2$ = E$ + "T07"
DL3$ = E$ + "X*9876543210xx#"
DL4$ = E$ + "R1252"
DL5$ = E$ + "T00"
DL6$ = E$ + "X*xExDxCxBxAxx#"
DL7$ = E$ + "E"
DwnLd$=DL1$+DL2$+DL3$+DL4$+DL5$+DL6$+DL7$
Print #1, DwnLd$
```

## **7. OMR-1102 EXTENDED COMMANDS**

In addition to the configuration download commands, there are a series of extended commands utilized to support configuration options.

Units shipped from the factory do not have any options enabled unless specified in the purchase or sales orders. Since OMR-1102 configuration options are factory set, they are not downloadable with application software but can be modified after shipment by working with the Customer Service Department. Configuration options are stored in non-volatile memory which means once a configuration option is set (enabled) it will remain active until it is disabled, even if the power is turned off.

The extended command set supports the following options:

- Document flow control
- Document exit control
- Timing track selection
- Data flow control
- Data mark location
- Parity format
- Retransmit
- Status
- Version

To implement an extended command, open the communications port and transmit the proper data string to the OMR-1102 to execute the dynamic features

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of various options. Three options do not have dynamic commands. They are data flow control, data mark location, and parity format.

The following is an example of syntax for extended command dynamics.

Required download format:

Ctrl R | ESC | Command [Argument] | Ctrl R | Command E

### **7.1 Document Feed Control**

**Action:** Places control of the OMR transport under the control of the application software allowing transport to be dynamically enabled or disabled.

**Syntax:** ESC#  
Where # is defined as follows:  
D = Disable  
E = Enable

**Remarks:** An operator prompt to insert a response form adequately controls timing for feeding response forms. However, in applications requiring extensive document processing, it may be desirable to disable the transport while the application software is processing the data string. The factory default for the document feed control option is set for control under the application program. Upon power up of the unit, the motor transport will be disabled. Consequently, if the response form is inserted and the application software does not enable the transport, the OMR will not accept the document. If an enable command is sent, the transport will remain enabled until a disable command is sent or a document is processed.

**Example:** The following example would enable the OMR transport:

```
Open COM1:9600,E as #1
ChrOut$=Chr$(18)+Chr$(27)+"E"+Chr$(18)+"E"
Print #1, ChrOut$
```

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## **7.2 Document Exit Control**

**Action:** Controls the response form exit after the data has been processed. The form exits out the rear of the OMR or returns to the operator.

**Syntax:** ESC##  
Where ## is defined as follows:  
ER = Exit Rear  
EF = Exit Front

**Remarks:** The response form is transported past the read head and then held in the reader. After data has been processed, the exit command is sent to continue the response form through the OMR and out the back, or through the entrance throat to the operator. Note: OMR logic will not accept additional commands while the processor is busy. Therefore, it is necessary all data be received by the host computer before the eject command is sent.

**Example:** The following example would eject the document out the front of the reader:

```
Open COM1:9600,E as #1
ChrOut$=Chr$(18)+Chr$(27)+"EF"+Chr$(18)+"E"
Print #1, ChrOut$
```

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### **7.3 Data Flow Control**

Action: Utilizes RTS/CTS or XON/XOFF to control data flow from the OMR to the application program.

Syntax: None required.

Remarks: Data flow control has no dynamic command and is a configuration option only. This feature is normally used in networked systems or when the OMR is mounted in a terminal where the host processor is in a multi-tasking environment and must respond to service interrupts. Receipt of data from the OMR is allowed only when the host is not busy with other functions.

When using the RTS/CTS feature, control is executed by controlling the CTS line from the host and the RTS line from the OMR. When CTS is held low the OMR will not transmit data or transport a response form. When CTS is held high the OMR will transport and transmit normally. The OMR will hold RTS low while transporting a document or transmitting data.

Note: When the flow control option is enabled coupled with the RTS/CTS feature, it is necessary to replace the Chatsworth Data cable with a standard Null modem cable.

The OMR responds normally to transporting and transmitting when it receives an XON and will continue processing documents until an XOFF is received. Similar to RTS/CTS, the OMR will not accept a document or transmit data when in the XOFF state.

The standard Chatsworth Data cable supplied with the OMR supports the XON/XOFF feature.

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## **7.4 Timing Track Selection**

Action: Selects the timing track channel.

Syntax: ESC S#  
Where # is defined as follows:  
T = Top timing (right-side as entering OMR)  
B = Bottom timing (left-side as entering OMR)

Remarks: In some applications, location of the timing track on the response form is critical to the application.

This option allows the OMR to look for a left-hand (bottom) or right-hand (top) timing track or a timing track on either the left or right-edge of the response form. However, a single response form can have only one timing track. If either top timing or bottom timing was specified for factory configuration, no dynamic command is required; the logic will automatically look for a top or bottom timing track.

Example: The following example looks for the timing track at the top of the response document:

```
Open COM1:9600,E as #1
ChrOut$=Chr$(18)+Chr$(27)+"ST"+Chr$(18)+"E"
Print #1, ChrOut$
```

## **7.5 Data Mark Alignment**

Action: Specifies data mark location in relationship to the timing track as either inline or offset.

Syntax: None required.

Remarks: Data mark location has no dynamic command (configuration option only). Data mark location with relationship to the timing track has two selections. 1) Inline timing specifies leading and trailing edges of the timing mark are within the response form defined mark area; 2) Offset timing specifies data mark area between the trailing edge of the lead timing mark and the leading edge of the following timing mark.

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## **7.6 Parity Format**

Action: Defines the output data string parity format for the OMR-1102

Syntax: None required.

Remarks: Parity format has no dynamic command (configuration option only). The default parity format is even, 7 bits and 1 stop bit. This format works on all standard P.C.'s and most controllers and the option allows for modification of parity format, if needed. Parity can be set to ODD/EVEN/NONE with 7 data bits or none with 8 data bits. The stop bit is always set at 1.

## **7.7 Retransmit**

Action: Retransmits the data buffer prior to processing the next document.

Syntax: ESC "R"

Remarks: The OMR normally reads and transmits the data in the specified format. However, certain conditions might make it necessary to retransmit the data.

Example: Open COM1:9600,E as #1  
ChrOut\$=Chr\$(18)+Chr\$(27)+"R"+Chr\$(18)+"E"  
Print #1, ChrOut\$

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## **7.8 OMR Status**

Action:      Retrieves the status byte from the OMR-1102.

Syntax:      ESC "S"

Remarks:    The OMR-1102 logic maintains a status byte. Each bit of the byte represents a specific condition of the unit. The definition of each bit is as follows:

|       |                            |
|-------|----------------------------|
| Bit 1 | Document at throat         |
| Bit 2 | Document jam               |
| Bit 3 | Power on self-test failure |
| Bit 4 | Not used                   |
| Bit 5 | Transport cleared          |
| Bit 6 | Transport enabled          |
| Bit 7 | Q.C. test mode             |
| Bit 8 | Parity bit                 |

Example:    OPEN COM1:9600,E as #1  
             ChrOut\$+Chr\$(18)+Chr\$(27)+"S"+Chr\$(18)+"E"  
             Print #1, ChrOut\$  
             Line Input #1, C\$  
             Locate 10,5:Print "OMR Status";ASC(C\$)  
             End

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## **7.9 Version Number**

Action: Transmit the version number of the EPROM and data information relative to the current program resident in the OMR-1102.

Syntax: ESC "V"

Remarks: The Version command transmits two character strings. The first string contains the model number of the unit followed by a space, then a dash, and another space, and the word "Version" and the version number.

Example: [OMR-1102 = Version X.XX]<CR>

The second string contains the date and time the firmware was generated. The format is as follows:

Day Of Week|Month|Day of Month|Year|Hour:Min.

Example: [WedNov23199416:23]<CR>

This information may help to diagnose a problem occurring during set-up or with the operation of the OMR; or to determine the version accommodates the software being utilized.

Example: Open COM1:9600,E as #1  
ChrOut\$=Chr\$(18)+Chr\$(27)+"V"+Chr\$(18)+"E"  
Print #1, ChrOut\$  
Line Input #1,C\$  
Locate 10,5:Print "Version Line 1";C\$  
Line Input #1,C\$  
Locate 12,5:Print "Version Line 2";C\$  
End

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## **8. UTILITY PROGRAMS**

The distribution disk supplied with the OMR-1102 contains four (4) utility programs designed to support specific areas. The utility programs are:

|               |                                                                     |
|---------------|---------------------------------------------------------------------|
| OMRCFG.EXE    | Configuration Update Program                                        |
| OMRDEMO.EXE   | Card Image Display                                                  |
| OMRDETECT.EXE | Detect OMR Com: Port<br>(including baud rate and cable information) |
| OMRDNLD.EXE   | OMR Universal Driver                                                |

### **8.1 OMRCFG.EXE**

Action: OMRCFG represents the OMR configuration (config) program designed to modify the default factory settings for baud rate and translation mode. The factory default is 9600 baud and binary translation.

Syntax: OMRCFG [/X/X]

Remarks: The config program is designed to execute factory defaults and COM1. However, if factory defaults are no longer set use the arguments /X to temporarily modify the program as follows:

/PX where X is a 1 for COM1: or 2 for COM2:

/BX where X is the current baud rate of the OMR (300-38400)

/Q where Q will cause the OMRCFG.CFG file to download to the OMR.

/? Where ? will cause the program to display the above options.

The config program will display a main menu as follows:

F1: Help  
F5: Restore factory defaults  
F7: Configure OMR-1102  
F10: Save and exit  
EXC: Abandon changes and exit

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When F7 is selected a sub-menu is displayed:

ComPort = COM1  
BaudRate = 9600  
Translation = Binary

Up and down arrows move through the list and the left and right arrows toggle to the default. The following options are available:

ComPort = COM1/COM2  
BaudRate = 300 to 38400  
Translation: Binary/Hollerith/True Data

Example: The following example will execute the program with the OMR1102 attached to the COM2: serial communications port.

```
C:>[CD\OMRUTIL] [Enter]
C:>OMRUTIL[OMRCFG/P2] [Enter]
```

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## **8.2 OMRDEMO.EXE**

- Action: Displays response document marks in graphic form.
- Syntax: OMRDEMO
- Remarks: OMRDEMO is used during checkout before beginning an application or in the development of software. This program also examines response forms to determine which marks the OMR detects as valid.
- Example: To execute the OMRDEMO program, log into the directory that contains the utility program and enter the following:
- ```
C:>[CD\OMRUTIL] [Enter]  
C:>OMRUTIL[OMRDEMO] [Enter]
```
- Output: A graphic displays response form rows horizontally and columns on the vertical axis. "*" indicates marks and "." indicates blanks. Row and column locations are aligned on the X/Y axis to assist in locating specific marks on the document.

8.3 OMRDETECT.EXE

- Action: Determines the current configuration of the OMR.
- Syntax: OMRDETECT
- Remarks: Utilizes machine code routines to determine configuration interface (attempts to identify which communications port, current baud rate, and if possible the data cable).
- Example: To execute OMRDETECT, log into the directory that contains the utility program. Execute the program by entering the program name.
- ```
C:>[CD\OMRUTIL] [Enter]
C:>OMRUTIL[OMRDETECT] [Enter]
```
- Output: An output message indicates if OMR was found, Com. Port in use, baud rate, and the cable type (if possible to determine).

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#### **8.4 OMRDNLD.EXE**

Action: Download interface program.

Syntax: OMRDNLD

Remarks: Utilizes command line subroutines allows a systems analyst or programmer to evaluate format and configuration download commands as they would apply to a specific response form.

Example: To execute OMRDNLD, log into the directory containing the utility program and execute the program by entering the program name.

```
C:>[CD\OMRUTIL] [Enter]
C:>OMRUTIL[OMRDNLD] [Enter]
```

Output: A visual displays various configuration download commands. Error messages/operator prompts assist in program usage.

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## **APPENDIX A**

### **DOCUMENT SPECIFICATIONS**

#### **1.0 SCOPE**

This document sets forth the design specification for cards scanned by the Chatsworth Data OMR-1102.

#### **2.0 PURPOSE**

This document provides accurate, uniform specifications in order to eliminate printing problems and therefore read errors.

#### **3.0 DIMENSIONS**

##### **3.1 Length**

The length can vary from 3.25 inches (82.55 mm) to 12.0 inches (304.8 mm).

##### **3.2 Width**

The width is 3.250 inches (82.55 mm)  $\pm$  0.010 inches (0.254 mm).

##### **3.3 Thickness**

Card thickness may be 0.0040 to 0.0080 inches (0.102 to 0.203 mm) for the OMR-1100 and OMR-9002. The OMR-2000 card escapement has been set for a 0.007 (0.178 mm) inch thick card; therefore, all media for the OMR-2000 must be 0.007 inch (0.178 mm) thick.

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## **4.0 PAPER STOCK**

### **4.1 Type**

A white 100% wood pulp paper; no ground wood allowed; total soft content not to exceed 40% shall be used. No smudges, water-marks, embossed or printed patterns, or fluorescent additives shall be present.

### **4.2 Basic Weight**

20 lb. bond or 50 lb. offset (75.2 g/sq. m) minimum  
40 lb. bond or 100 lb. offset (150.4 g/sq. m) maximum

### **4.3 Reflectance**

The entire stock should measure 75% reflectance (minimum) in the red to near infrared region. For offset stock, the reading surface of the document must be the smooth surface.

Note: Reflectance is to be measured on a Moore Model 082 Reflectometer or equivalent, using a barium sulfate plaque as a 100% standard reference.

### **4.4 Dirt in Paper**

The dirt content shall not exceed 10 parts per million.

### **4.5 Smoothness**

The smoothness of paper is a measure of the time required for a fixed volume of air to leak across the surface of a paper sample. Paper smoothness shall be between 100 and 200 using the Sheffield method.

### **4.6 Porosity**

The definition of porosity is the resistance of paper to the passage of air under a specified pressure through the paper. Porosity of paper as measured by the Gurley test method shall be between 10 and 15.

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#### **4.7 Grain**

In order to reduce the effect of humidity on the form the grain of the paper should be cut parallel to the long direction, “grain long.”

Note: This is a very important but often misunderstood parameter. Paper will expand when damp due to humidity. The expansion is greatest along the “long grain” of the paper. Cutting the paper with the grain parallel to the direction of paper feed minimizes the expansion of the width of the media, reducing the potential for jamming.

#### **4.8 Tear Resistance**

This is defined as the average force in grams required to tear a single sheet of paper after the tear has been started. The paper must withstand 40 to 70 grams (Elmendorf) in either direction.

#### **4.9 Curl**

The maximum allowable curl of the card measured from a flat surface on which a card is laying to the most protruding point on the card is 0.05 inches (1.27 mm).

### **5.0 PRINTING**

In the active data section of all cards, print will be in a reflective (red in the case of reader containing the ink read head option) or black non-reflective, as specified in Section 6 below. For the non-data sections of the card (i.e. advertising or instructional sections), inks of all colors can be used. However, no advertising or instructional printing is allowed on the edge containing the strobe area. Additionally, the leading 0.125 inches (3.14 mm) shall be clear of printing in all channels to facilitate proper detection of the media.

#### **5.1 Pre-printed Marks and Strobe Marks**

The ink used for all strobe marks and pre-printed data marks will be of a non-reflective black type having a reflectance measurement of less than 10%.

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## **5.2 Background Printing**

The ink used in printing formatted background information on the card which is intended not to be read by the OMR shall be a reflective type having a reflectance measurement greater than 80% when measured on a Moore Model 082 Reflectometer. The following PMS inks may be used:

|     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 101 | 102 | 106 | 107 | 108 | 113 | 114 | 115 | 116 | 120 | 121 |
| 122 | 123 | 127 | 128 | 129 | 130 | 134 | 135 | 136 | 137 | 141 |
| 142 | 143 | 144 | 148 | 149 | 150 | 151 | 155 | 156 | 157 | 158 |
| 162 | 163 | 164 | 165 | 169 | 170 | 171 | 172 | 176 | 177 | 178 |
| 182 | 183 | 184 | 185 | 189 | 190 | 191 | 192 | 196 | 197 | 198 |
| 203 | 204 | 205 | 210 | 211 | 212 | 217 | 218 | 219 | 223 | 224 |
| 225 | 230 | 231 | 232 | 236 | 237 | 238 | 243 | 244 | 245 | 250 |
| 251 | 252 | 485 | 486 | 487 | 488 | 489 |     |     |     |     |

Note: If the reader has the ink read head option, a red ink must be used. A suitable red ink is PMS 177.

## **6.0 FORMAT**

### **6.1 Strobe Marks**

The strobe marks shall be printed along the bottom edge of the card with the leading edge of the card to the left when viewing the card face up.

The height of the strobe mark (measured across the width of the card) shall be  $0.125 \pm 0.005$  inches ( $3.127 \pm 0.127$  mm). The space between strobe marks (not center to center) shall not be less than 0.050 inches (1.27 mm). The strobe mark thickness (measured along the length of the card) must be a minimum of 0.030 inches (0.762 mm). The minimum distance from the leading edge of the card to the first strobe mark is 0.250 inches (6.35 mm). The minimum distance from the last strobe mark to the trailing edge of the card is also 0.250 inches (6.35 mm).

In the design it should be noted that the data read area exactly coincides with the length of the strobe mark. Therefore, strobe marks printed in line with the data marking area shall be the same width as, or wider than (no less than 95%) the data marking field.

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## **6.2 Data Areas**

There are exactly 12 possible data marking areas (channels) across the width of one column of the card. These marking areas are located exactly above the strobe mark. The centerline of the bottom row (above the strobe mark) is located at  $0.250 \pm 0.005$  inches ( $6.35 \pm 0.13$  mm) from the bottom edge of the card. The centerlines of all 12 rows are located at  $0.250 \pm 0.005$  ( $6.35 \pm 0.13$  mm) increments. The top row centerline is also located at  $0.250 \pm 0.005$  inches from the top edge of the card. The increment tolerance, 0.005 (0.13 mm), is non-accumulative.

The maximum height of a marking area shall be 0.215 inch (5.461 mm). The minimum height of a marking data area is 0.125 inch (3.175 mm).

## **6.3 Instructional/Advertising Area**

This area may be used as the user chooses. However, it is important to note that if colors other than the designated reflective and black are used, the product is more expensive and few printers can provide accurate forms.

## **7.0 ENVIRONMENT**

The cards should be packed in a manner that protects them from folding and damaging their edges. The packaging should be environmentally resistant (that is in cellophane, etc.) to protect the cards from excessive humidity and moisture. The sealed but not necessarily airtight package should contain no more than 1000 cards. The cards should be stored in an area with a free flow of air.

### **7.1 Storage**

Forms must not be stored directly on the floor or exposed to condensation. The environment conditions must be as follows:

Temperature: 0° C to 60° C Ambient

Humidity: 20% to 80% Relative Humidity, non-condensing

### **7.2 Operating**

0° C to 55° C at 20% to 80% Relative Humidity

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## **APPENDIX B**

### **FACTORY DEFAULTS**

|                         |                    |
|-------------------------|--------------------|
| Baud Rate:              | 9600               |
| Parity:                 | Even               |
| Data Mark Alignment:    | Inline             |
| Timing Track Selection: | Bottom (Left Side) |
| Data Mark Conversion:   | Binary ASCII       |
| Row Enable:             | All Rows Enabled   |
| Document Feed Control:  | Feed Through       |
| Data Flow Control:      | None               |

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## APPENDIX C

### HOLLERITH TO ASCII CONVERSION

| ASCII |     |      | HOLLERITH   |
|-------|-----|------|-------------|
| DEC   | OCT | CHAR |             |
| 0     | 000 | NUL  | 9-8-1-0-12  |
| 1     | 001 | SOH  | 9-1-12      |
| 2     | 002 | STX  | 9-2-12      |
| 3     | 003 | ETX  | 9-3-12      |
| 4     | 004 | EOT  | 9-7         |
| 5     | 005 | ENQ  | 9-8-5-0     |
| 6     | 006 | ACK  | 9-8-6-0     |
| 7     | 007 | BEL  | 9-8-7-0     |
| 8     | 010 | BS   | 9-6-11      |
| 9     | 011 | HT   | 9-5-12      |
| 10    | 012 | LF   | 9-5-0       |
| 11    | 013 | VT   | 9-8-3-12    |
| 12    | 014 | FF   | 9-8-4-12    |
| 13    | 015 | CR   | 9-8-5-12    |
| 14    | 016 | SO   | 9-8-6-12    |
| 15    | 017 | SI   | 9-8-7-12    |
| 16    | 020 | DLE  | 9-8-1-11-12 |
| 17    | 021 | DC1  | 9-1-11      |
| 18    | 022 | DC2  | 9-2-11      |
| 19    | 023 | DC3  | 9-3-11      |
| 20    | 024 | DC4  | 9-8-4       |
| 21    | 025 | NAK  | 9-8-5       |
| 22    | 026 | SYN  | 9-2         |
| 23    | 027 | ETB  | 9-6-0       |
| 24    | 030 | CAN  | 9-8-11      |
| 25    | 031 | EM   | 9-8-1-11    |
| 26    | 032 | SUB  | 9-8-7       |
| 27    | 033 | ESC  | 9-7-0       |

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| ASCII |     |      | HOLLERITH |
|-------|-----|------|-----------|
| DEC   | OCT | CHAR |           |
| 28    | 034 | FS   | 9-8-4-11  |
| 29    | 035 | GS   | 9-8-5-11  |
| 30    | 036 | RS   | 9-8-6-11  |
| 31    | 037 | US   | 9-8-7-11  |
| 32    | 040 | SP   | (Blank)   |
| 33    | 041 | !    | 8-7-12    |
| 34    | 042 | "    | 8-7       |
| 35    | 043 | #    | 8-3       |
| 36    | 044 | \$   | 8-3-11    |
| 37    | 045 | %    | 8-4-0     |
| 38    | 046 | &    | 12        |
| 39    | 047 | '    | 8-5       |
| 40    | 050 | (    | 8-5-12    |
| 41    | 051 | )    | 8-5-11    |
| 42    | 052 | *    | 8-4-11    |
| 43    | 053 | +    | 8-6-12    |
| 44    | 054 | ,    | 8-3-0     |
| 45    | 055 | -    | 11        |
| 46    | 056 | .    | 8-3-12    |
| 47    | 057 | /    | 1-0       |
| 48    | 060 | 0    | 0         |
| 49    | 061 | 1    | 1         |
| 50    | 062 | 2    | 2         |
| 51    | 063 | 3    | 3         |
| 52    | 064 | 4    | 4         |
| 53    | 065 | 5    | 5         |
| 54    | 066 | 6    | 6         |
| 55    | 067 | 7    | 7         |
| 56    | 070 | 8    | 8         |
| 57    | 071 | 9    | 9         |
| 58    | 072 | :    | 8-2       |
| 59    | 073 | ;    | 8-6-11    |
| 60    | 074 | <    | 8-4-12    |
| 61    | 075 | =    | 8-6       |
| 62    | 076 | >    | 8-6-0     |
| 63    | 077 | ?    | 8-7-0     |
| 64    | 100 | @    | 8-4       |

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| ASCII |     |       | HOLLERITH |
|-------|-----|-------|-----------|
| DEC   | OCT | CHAR  |           |
| 65    | 101 | A     | 1-12      |
| 66    | 102 | B     | 2-12      |
| 67    | 103 | C     | 3-12      |
| 68    | 104 | D     | 4-12      |
| 69    | 105 | E     | 5-12      |
| 70    | 106 | F     | 6-12      |
| 71    | 107 | G     | 7-12      |
| 72    | 110 | H     | 8-12      |
| 73    | 111 | I     | 9-12      |
| 74    | 112 | J     | 1-11      |
| 75    | 113 | K     | 2-11      |
| 76    | 114 | L     | 3-11      |
| 77    | 115 | M     | 4-11      |
| 78    | 116 | N     | 5-11      |
| 79    | 117 | O     | 6-11      |
| 80    | 120 | P     | 7-11      |
| 81    | 121 | Q     | 8-11      |
| 82    | 122 | R     | 9-11      |
| 83    | 123 | S     | 2-0       |
| 84    | 124 | T     | 3-0       |
| 85    | 125 | U     | 4-0       |
| 86    | 126 | V     | 5-0       |
| 87    | 127 | W     | 6-0       |
| 88    | 130 | X     | 7-0       |
| 89    | 131 | Y     | 8-0       |
| 90    | 132 | Z     | 9-0       |
| 91    | 133 | [     | 8-2-12    |
| 92    | 134 | \     | 8-2-0     |
| 93    | 135 | ]     | 8-2-11    |
| 94    | 136 | ^ (↑) | 8-7-11    |
| 95    | 137 | — (←) | 8-5-0     |
| 96    | 140 | — `   | 8-1       |
| 97    | 141 | a     | 1-0-1     |
| 98    | 142 | b     | 2-0-12    |
| 99    | 143 | c     | 3-0-12    |

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| ASCII |     |      | HOLLERITH |
|-------|-----|------|-----------|
| DEC   | OCT | CHAR |           |
| 100   | 144 | d    | 4-0-12    |
| 101   | 145 | e    | 5-0-12    |
| 102   | 146 | f    | 6-0-12    |
| 103   | 147 | g    | 7-0-12    |
| 104   | 150 | h    | 8-0-12    |
| 105   | 151 | i    | 9-0-12    |
| 106   | 152 | j    | 1-11-12   |
| 107   | 153 | k    | 2-11-12   |
| 108   | 154 | l    | 3-11-12   |
| 109   | 155 | m    | 4-11-12   |
| 110   | 156 | n    | 5-11-12   |
| 111   | 157 | o    | 6-11-12   |
| 112   | 160 | p    | 7-11-12   |
| 113   | 161 | q    | 8-11-12   |
| 114   | 162 | r    | 9-11-12   |
| 115   | 163 | s    | 2-0-11    |
| 116   | 164 | t    | 3-0-11    |
| 117   | 165 | u    | 4-0-11    |
| 118   | 166 | v    | 5-0-11    |
| 119   | 167 | w    | 6-0-11    |
| 120   | 170 | x    | 7-0-11    |
| 121   | 171 | y    | 8-0-11    |
| 122   | 172 | z    | 9-0-11    |
| 123   | 173 | {    | 0-12      |
| 124   | 174 |      | 11-12     |
| 125   | 175 | }    | 0-11      |
| 126   | 176 | ~    | 1-0-11    |
| 127   | 177 | DEL  | 9-7-12    |

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## APPENDIX D

### BINARY TABLE

|                |                |                |                |                |                |                | →  | 0  | 0 | 1 | 1   |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----|----|---|---|-----|
|                |                |                |                |                |                |                | →  | 1  | 1 | 1 | 1   |
|                |                |                |                |                |                |                | →  | 0  | 1 | 0 | 1   |
| b <sub>7</sub> | b <sub>6</sub> | b <sub>5</sub> | b <sub>4</sub> | b <sub>3</sub> | b <sub>2</sub> | b <sub>1</sub> |    | 1  | 2 | 3 | 4   |
|                |                |                | 0              | 0              | 0              | 0              | 0  | SP | 0 | ` | p   |
|                |                |                | 0              | 0              | 0              | 1              | 1  | !  | 1 | a | q   |
|                |                |                | 0              | 0              | 1              | 0              | 2  | "  | 2 | b | r   |
|                |                |                | 0              | 0              | 1              | 1              | 3  | #  | 3 | c | s   |
|                |                |                | 0              | 1              | 0              | 0              | 4  | \$ | 4 | d | t   |
|                |                |                | 0              | 1              | 0              | 1              | 5  | %  | 5 | e | u   |
|                |                |                | 0              | 1              | 1              | 0              | 6  | &  | 6 | f | v   |
|                |                |                | 0              | 1              | 1              | 1              | 7  | '  | 7 | g | w   |
|                |                |                | 1              | 0              | 0              | 0              | 8  | (  | 8 | h | x   |
|                |                |                | 1              | 0              | 0              | 1              | 9  | )  | 9 | i | y   |
|                |                |                | 1              | 0              | 1              | 0              | 10 | *  | : | j | z   |
|                |                |                | 1              | 0              | 1              | 1              | 11 | +  | ; | k | {   |
|                |                |                | 1              | 1              | 0              | 0              | 12 | ,  | < | l |     |
|                |                |                | 1              | 1              | 0              | 1              | 13 | -  | = | m | }   |
|                |                |                | 1              | 1              | 1              | 0              | 14 | .  | > | n | ~   |
|                |                |                | 1              | 1              | 1              | 1              | 15 | /  | ? | o | DEL |

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## **APPENDIX E**

### **SERVICE AND TECHNICAL SUPPORT**

In the event that your OMR-1102 requires service or repair, Chatsworth Data Corporation provides the following options:

- |                  |                                                                   |
|------------------|-------------------------------------------------------------------|
| Standard Repair: | OMR is repaired and reshipped within 7 working days.              |
| Priority Repair: | OMR is repaired and reshipped within 24 hours.                    |
| Exchange:        | Reader is replaced with another (when available) within 24 hours. |

Readers returned to Chatsworth Data for service/repair require an RMA (Return Merchandise Authorization) number. When contacting Customer Service for the RMA, please provide the model and serial numbers located on the identification label on the bottom of the OMR. Please return the reader in the original container, if possible, or make sure it is adequately packaged to insure no damage in transit.

#### **Contact and Shipping Information:**

Customer Service and Technical Support  
818-341-9200 (tel.)  
818-341-3002 (fax)  
800-423-5217 (outside CA)

Chatsworth Data Corporation  
20710 Lassen Street  
Chatsworth, CA 91311